This procedure is applicable to below models:

V810 STD	✓	S1	✓	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	06 July 2017	First Release	Chin Chee Yan

Content

Removal Procedure

1. Unload board from machine if any.
2. Go to X ray source tab and toggle “Turn off X-Ray”. Power off at Operator Console Panel

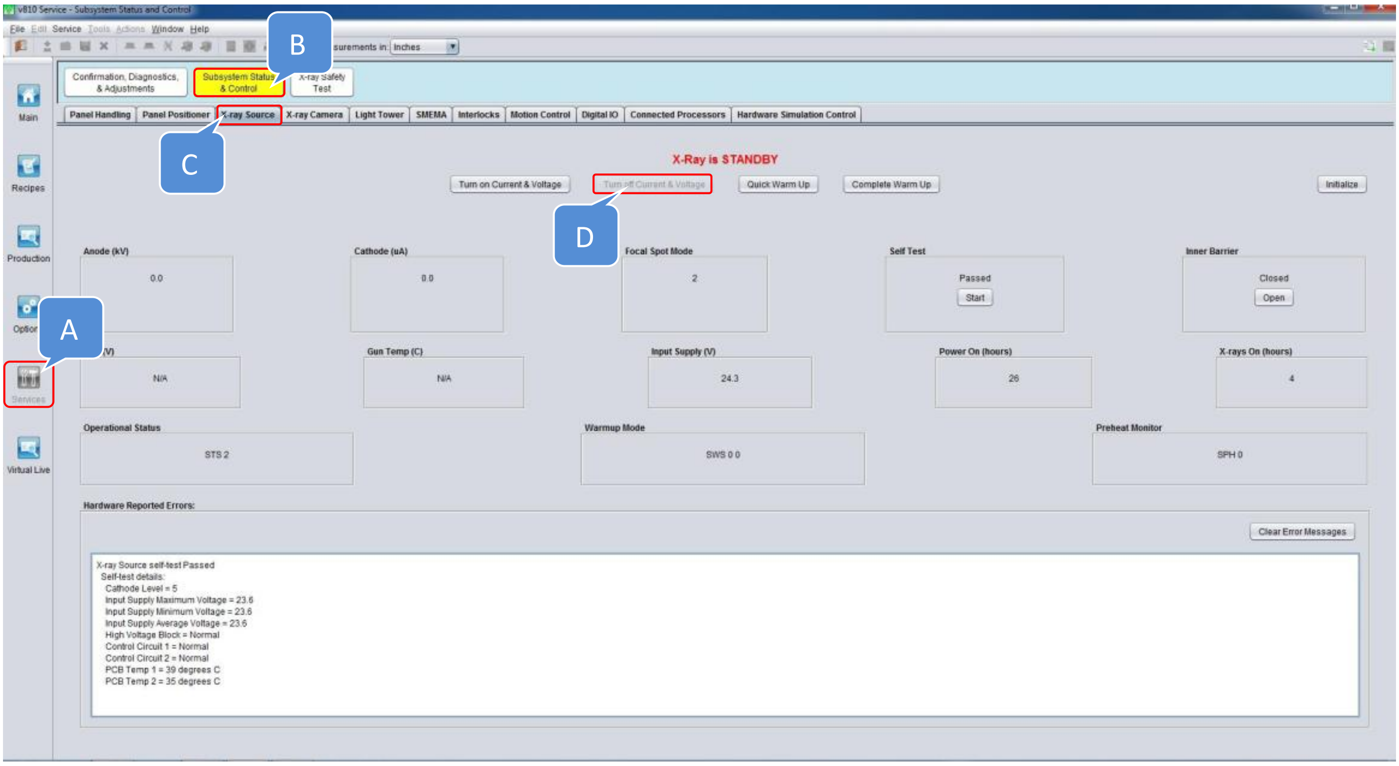

Figure 1 X Ray Source Shutdown

3. Power off SSCA

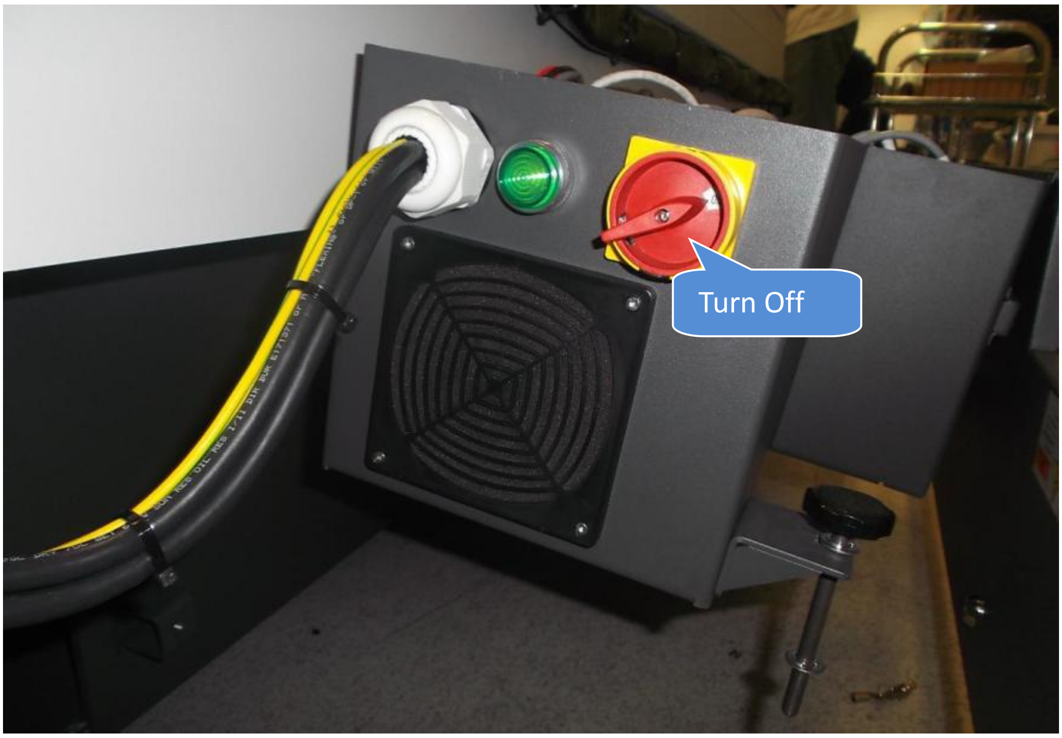

Figure 2 SSCA Power Off

4. Gently loosen Both Anchors as below Figure 3

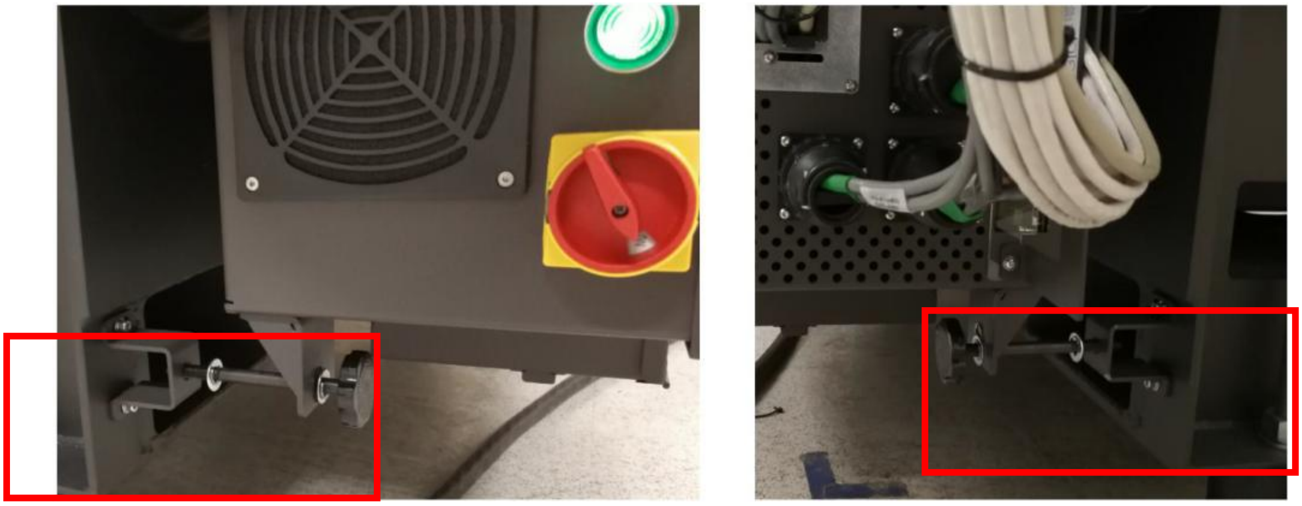

Figure 3 Anchors Release

5. Extend SSCA Support Rod which locate underneath of SSCA as below Figure 4

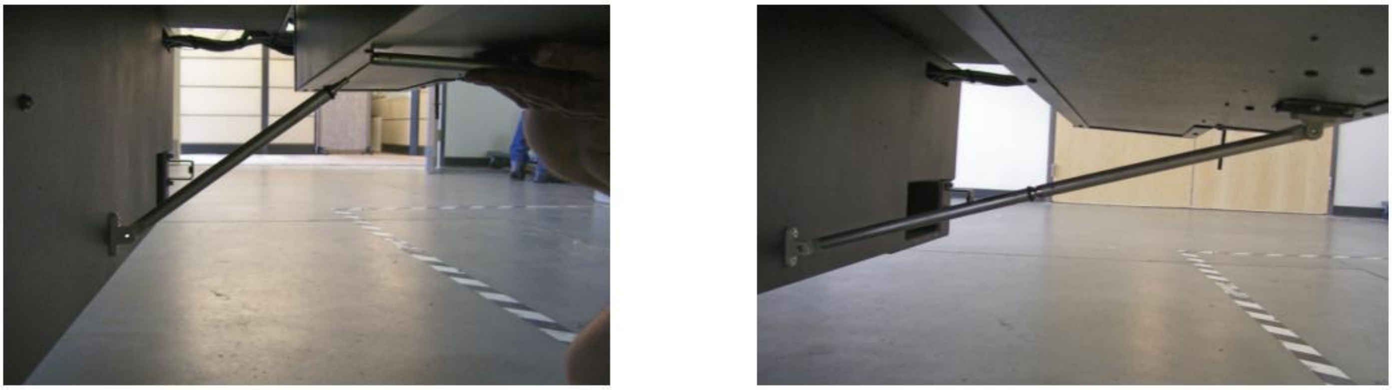

Figure 4 SSCA Support Rod Extend

I/O, MODULE, SYNQNET NETWORK ADAPTER Removal

6. Unlock Slice I/O, MODULE,SYNQNET NETWORK ADAPTER as below Figure 5

   Remarks :Unlock side slice as well where mainly ease of removal process

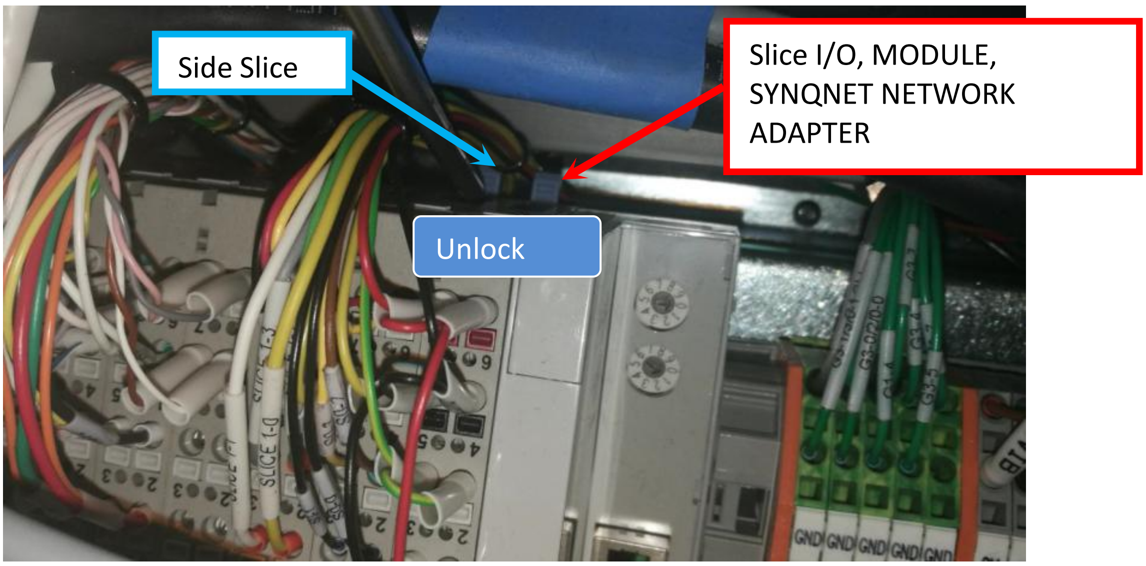

Figure 5 Slice (SYNQNET NETWORK ADAPATER) Unlock

7. Probe Tiny Square Button with Flat Head Screwdriver to disconnect IO Cables.

   Remarks : Perform Labeling at IO Cable upon removal. Applicable only for Cable Without Label

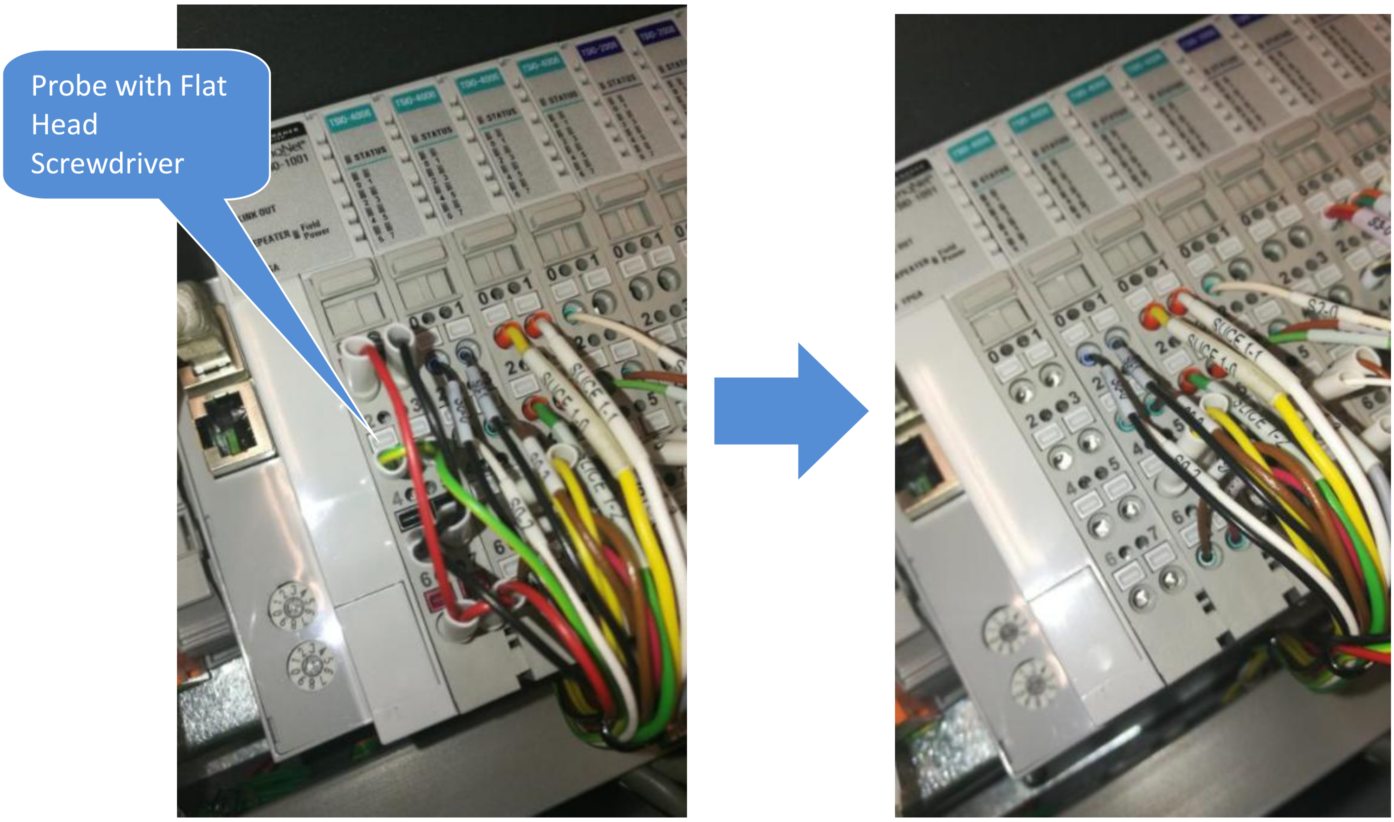

Figure 6 IO Cable Disconnect

8. Disconnect LAN cable as shown.

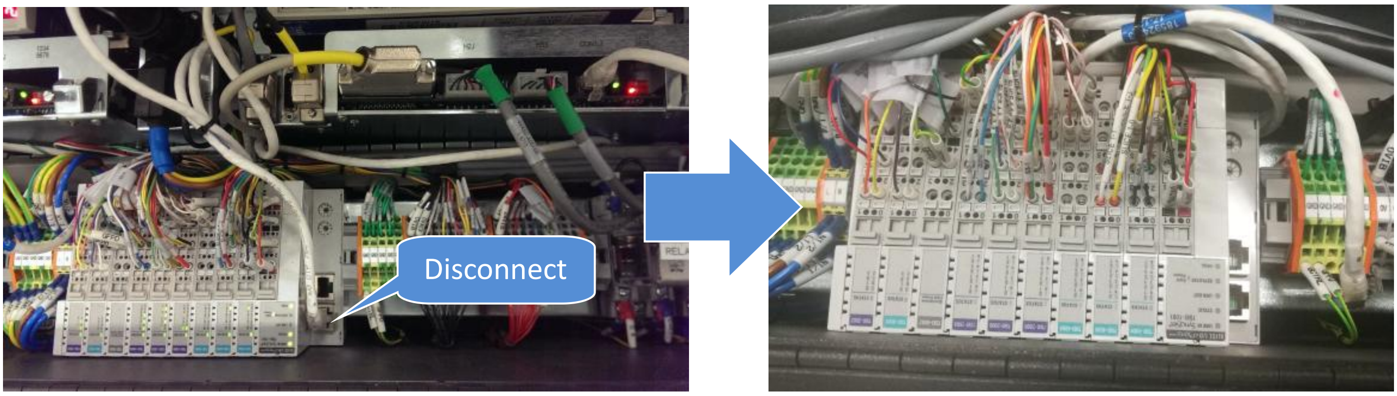

Figure 7 LAN cable Disconnect

9. Access from Red Arrow Indicated. Gently pull Slice out from IO Module

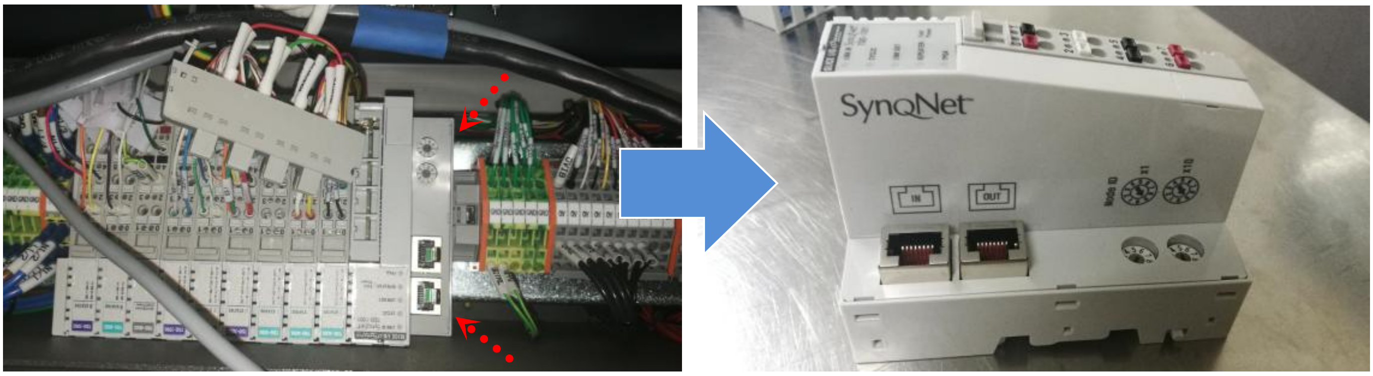

Figure 8 I/O, MODULE, SYNQNET NETWORK ADAPTER Removal

Selected Slice IO Removal

10. Unlock the lever as shown.

Remarks :Unlock side slices as well where mainly ease of removal process

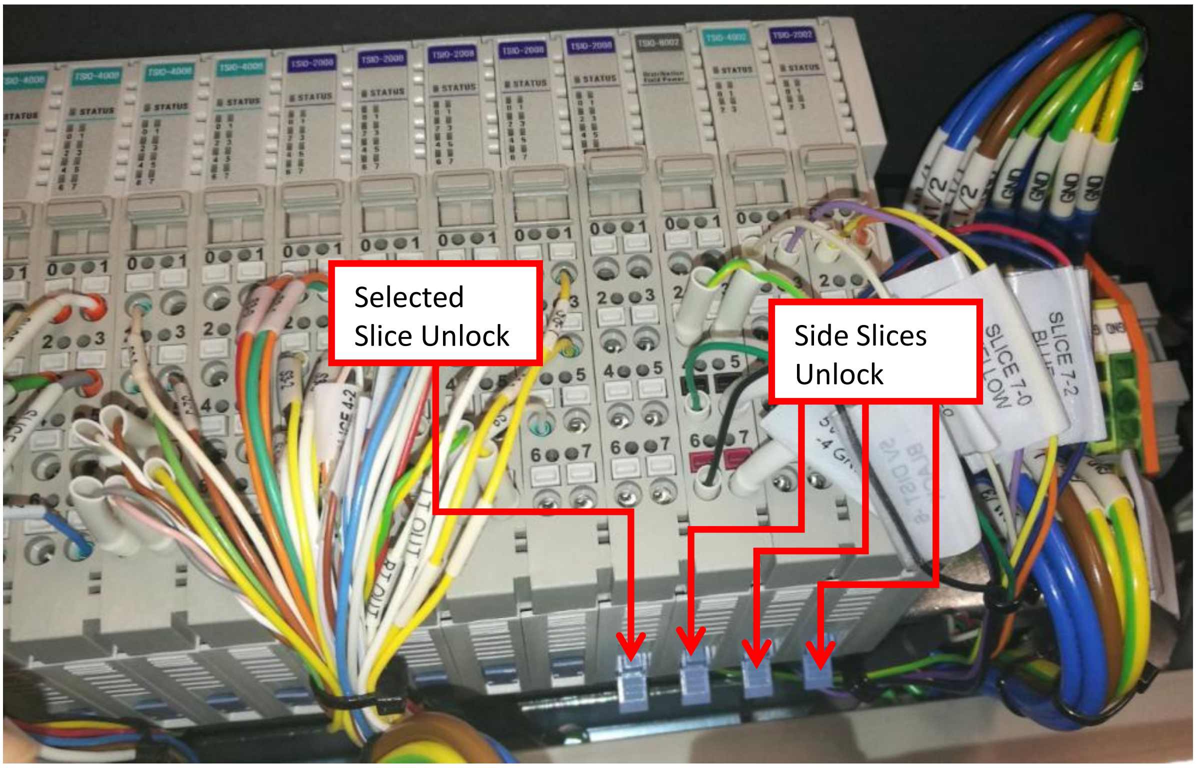

Figure 9 Selected Slice Unlock

11. Gently probe the Tiny Square Button at Selected Slice IO. Disconnect all IO Cable from Selected Slice IO

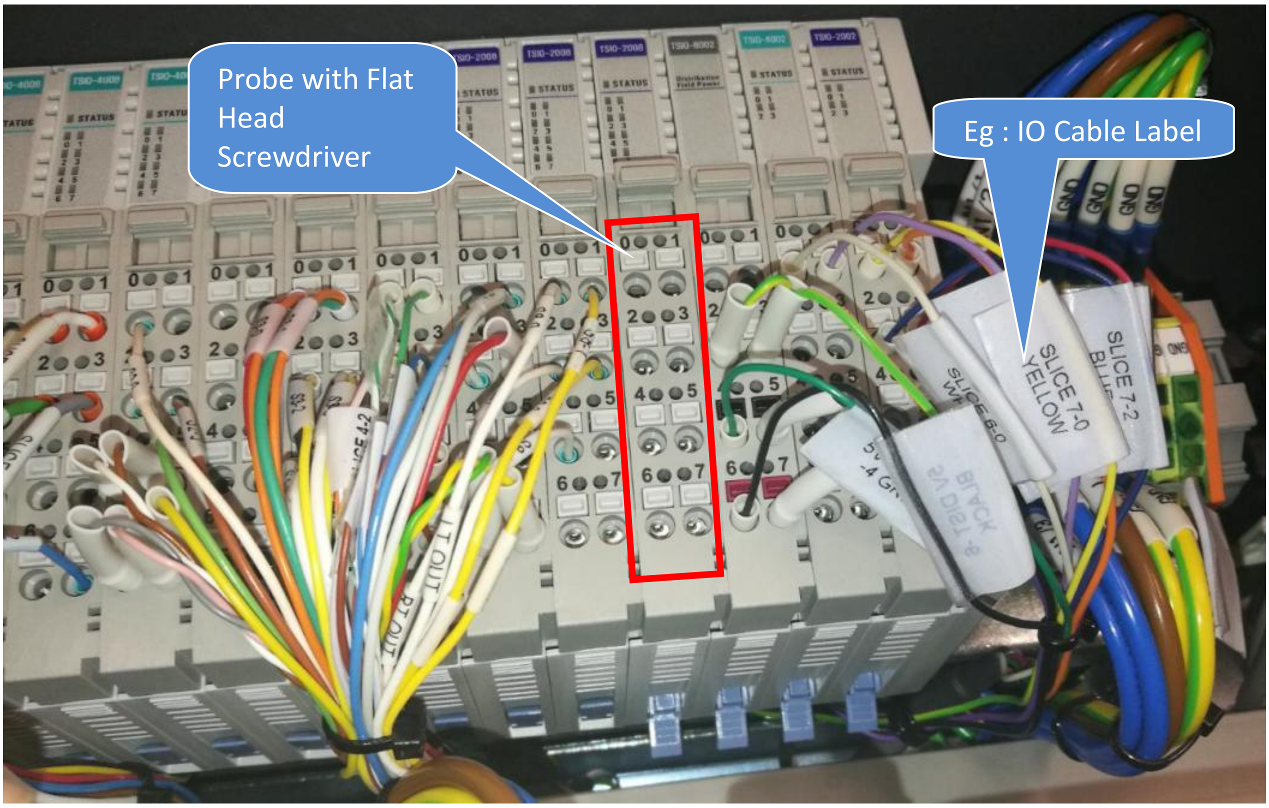

Figure 10 Selected IO Cable Removal

12. Remove the IO Module Side cover as shown.

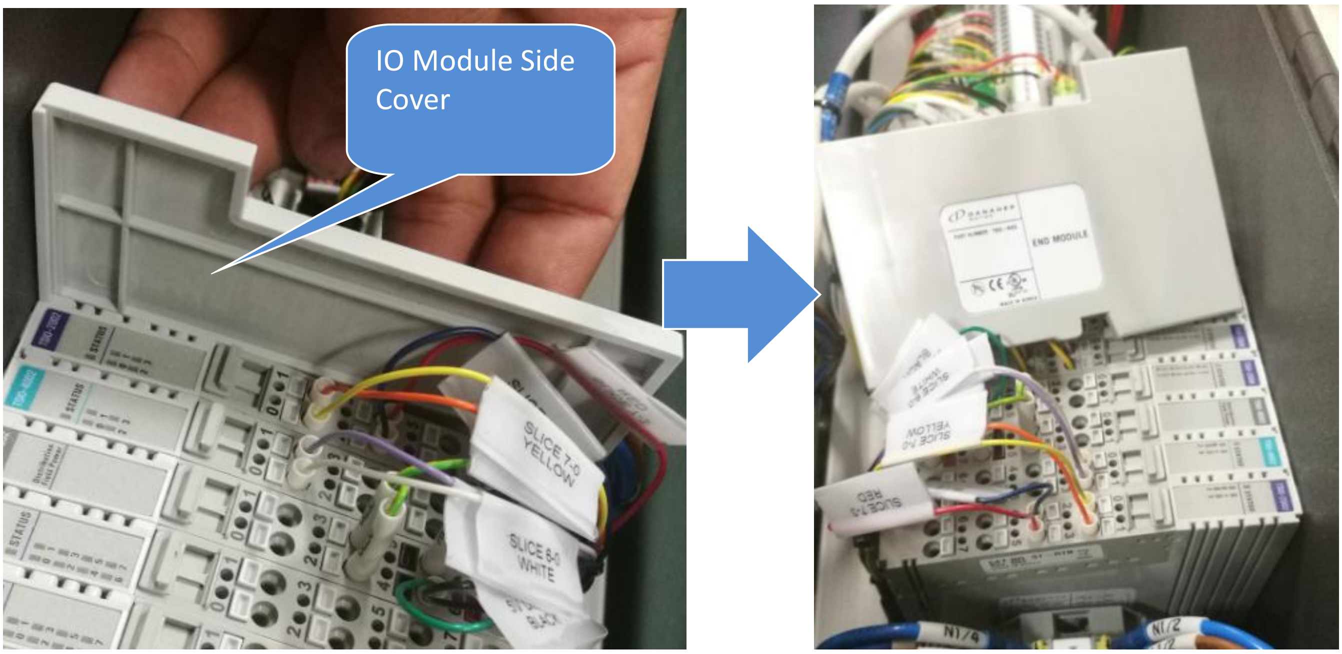

Figure 11 IO Module Side Cover Removal

13. Access from the most right side. Remove All Slice IO Slice as below Figure 12

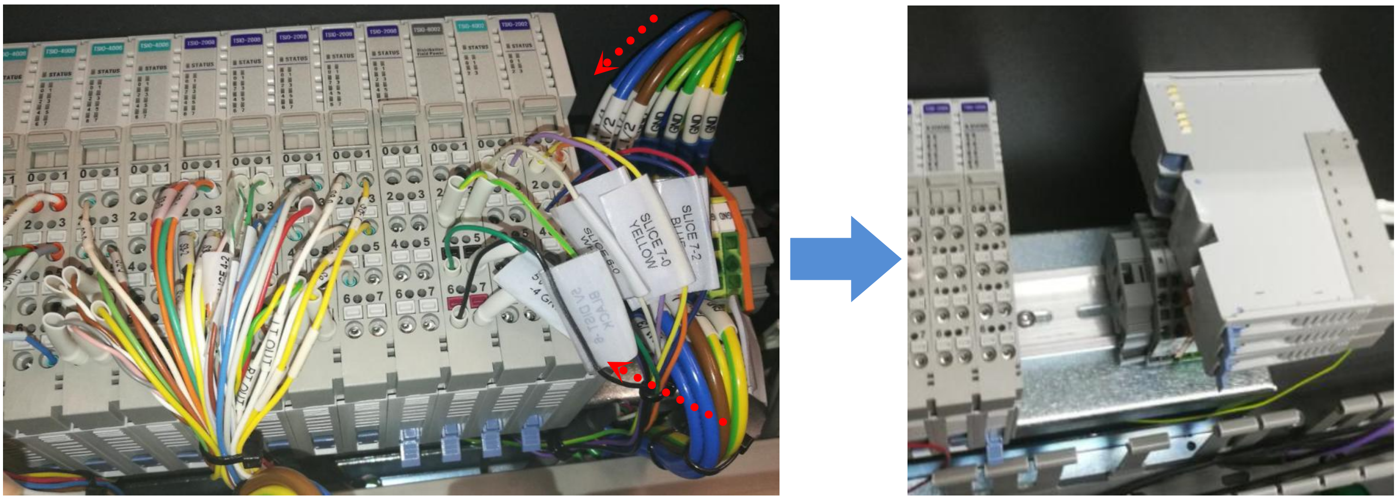

Figure 12 Side IO Slice Removal (Reference Only)

14. Remove the Selected Slice IO out from IO Module

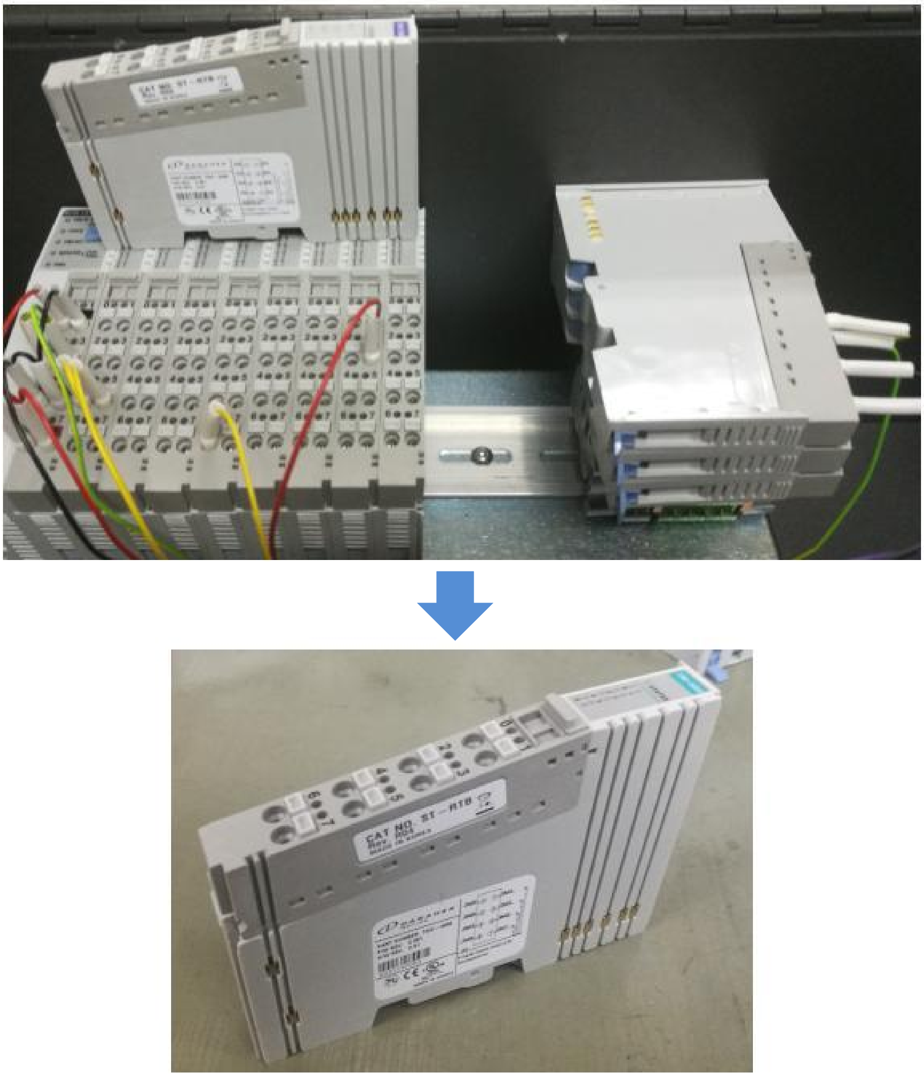

Figure 13 Selected Slice IO Removal

Replacement Procedure

15. AXI V810 STD machine DIO Signal Pin Layout is illustrated as below.

Standart																			
7-7	7-6	6-7	6-6	6	7	5-7	5-6	4-7	4-6	3-7	3-6	2-7	2-6	1-7	1-6	0-7	0-6	6	7
DIO In - 31	DIO In - 30	DIO Out - 31	DIO Out - 30			DIO In - 23	DIO In - 22	DIO In - 15	DIO In - 14	DIO In - 7	DIO In - 6	DIO Out - 23	DIO Out - 22	DIO Out - 15	DIO Out - 14	DIO Out - 7	DIO Out - 6		
				5V	5V	SMEMA Conveyor Panel Available	SMEMA Conveyor Not Ready	Interlock Service Mode On	Interlock Powered On	R.PIP Sensor Engaged	L.PIP Sensor Engaged	SHEMA Untested Panel Out	SHEMA Good Panel Out		3/2 Solenoid Valve VM (Brake)	Innner Barrier Control Valve	Panel Clamps Open/Close	24V	24V
7-5	7-4	6-5	6-4	4	5	5-5	5-4	4-5	4-4	3-5	3-4	2-5	2-4	1-5	1-4	0-5	0-4	4	5
DIO In - 29	DIO In - 28	DIO Out - 29	DIO Out - 28			DIO In - 21	DIO In - 20	DIO In - 13	DIO In - 12	DIO In - 5	DIO In - 4	DIO Out - 21	DIO Out - 20	DIO Out - 13	DIO Out - 12	DIO Out - 5	DIO Out - 4		
				0V	0V	Right VM Filter Sensor	Left VM Filter Sensor	Interlock Logic Bit #5	Interlock Logic Bit #4	R.Panel Clear Sensor Clear	L.Panel Clear Sensor Clear	SHEMA Tester Panel Available	SHEMA Tester Not Ready	5/2 Solenoid Valve	Stage Light On/Off	Belt Motor L/R	Belt Motor On/Off	0V	0V
7-3	7-2	6-3	6-2	2	3	5-3	5-2	4-3	4-2	3-3	3-2	2-3	2-2	1-3	1-2	0-3	0-2	2	3
DIO In - 27	DIO In - 26	DIO Out - 27	DIO Out - 26			DIO In - 19	DIO In - 18	DIO In - 11	DIO In - 10	DIO In - 3	DIO In - 2	DIO Out - 19	DIO Out - 18	DIO Out - 11	DIO Out - 10	DIO Out - 3	DIO Out - 2		
Cameras Controller Phase Locked	Cameras Not Using Excessive Current			GND	GND	Cylinder Bottom Sensor (VM)	Cylinder Top Sensor (VM)	Interlock Logic Bit #3	Interlock Logic Bit #2	Right Outer Barrier Open	Left Outer Barrier Open			Light Tower Buzzer	Light Tower Green	Right PIP Extended	Left PIP Extended	GND	GND
7-1	7-0	6-1	6-0	1	0	5-1	5-0	4-1	4-0	3-1	3-0	2-1	2-0	1-1	1-0	0-1	0-0	1	0
DIO In - 25	DIO In - 24	DIO Out - 25	DIO Out - 24			DIO In - 17	DIO In - 16	DIO In - 9	DIO In - 8	DIO In - 1	DIO In - 0	DIO Out - 17	DIO Out - 16	DIO Out - 9	DIO Out - 8	DIO Out - 1	DIO Out - 0		
Camera Controller Power On	Camera Control Cables Connected	Cameras Control Spare	Cameras Synth Clock			Inner Barrier Open	Inner Barrier Closed	Interlock Logic Bit #1	Interlock Logic Bit #0	Right Outer Barrier Closed	Left Outer Barrier Closed			Light Tower Yellow	Light Tower Red	Right Outer Barrier Control Valve	Left Outer Barrier Control Valve	0V	24V
Slice 7 - Input		Slice 6 - Output		Field Power Distributor		Slice 5 - Input		Slice 4 - Input		Slice 3 - Input		Slice 2 - Output		Slice 1 - Output		Slice 0 - Output		MES Slice IO Controller	

Figure 14 DIO Signal Pin Block Layout

16. Table 1 is indicate AXI STD machine Slide IO Part Number (Red Column at Figure 14)

No	Slide	Part Number	Description
1	MES Slice IO Controller	2150-0020	I/O, MODULE, SYNQNET NETWORK ADAPTER (TSIO-1001)
2	Slice 0 - Output	2150-0021	I/O, MODULE, 8-CHANNEL +24V OUTPUT (TSIO-4006)
3	Slice 1 - Output	2150-0021	I/O, MODULE, 8-CHANNEL +24V OUTPUT (TSIO-4006)
4	Slice 2 - Output	2150-0021	I/O, MODULE, 8-CHANNEL +24V OUTPUT (TSIO-4006)
5	Slice 3 - Input	2150-0022	I/O, MODULE, 8-CHANNEL +24V INPUT (TSIO-2008)
6	Slice 4 - Input	2150-0022	I/O, MODULE, 8-CHANNEL +24V INPUT (TSIO-2008)
7	Slice 5 - Input	2150-0022	I/O, MODULE, 8-CHANNEL +24V INPUT (TSIO-2008)
8	Field Power Distributor	2140-0005	POWER SUPPLY, SILECE I/O DISTRIBUTOR (TSIO-8002)
9	Slice 6 - Output	2150-0023	I/O, MODULE, 4-CHANNEL +5V OUTPUT (TSIO-4002)
10	Slice 7 - Input	2150-0024	I/O, MODULE, 4-CHANNEL +5V INPUT (TSIO-2002)

Table 1 Slice I/O Part number

I/O, MODULE, SYNQNET NETWORK ADAPTER Replacement

17. Prepare new I/O, MODULE, SYNQNET NETWORK ADAPTER (TSIO-1001) (P/N, 2150-0020) and unlock Lever as below Figure 15

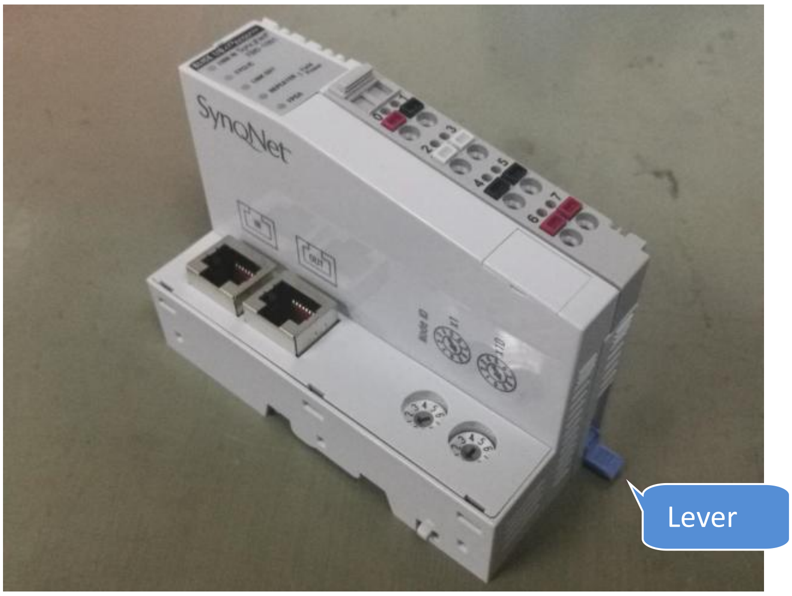

Figure 15 I/O, MODULE, SYNQNET NETWORK ADAPTER (TSIO-1001)

18. Replace new 2150-0020 by refer to below orientation. Slot on the track & gently push in & lock the lever accordingly.

    Remarks : Lock the Side Slice IO as well if unlock previously

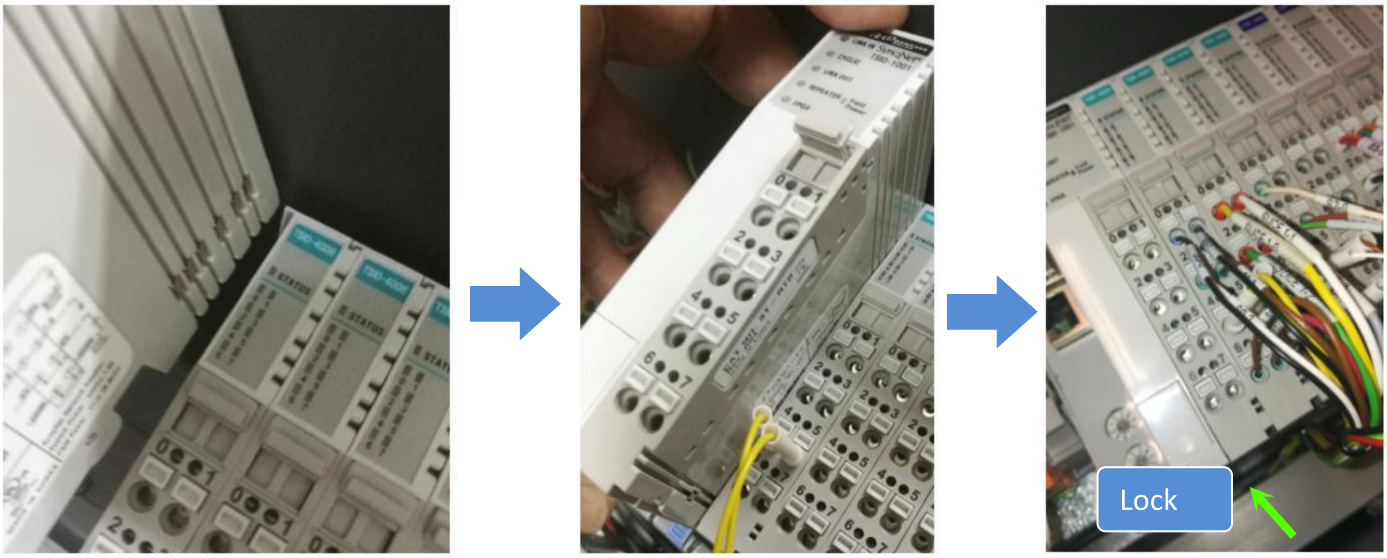

Figure 16

19. While probe the Tiny Square Button, gently insert IO Cable respectively by refer to Figure 14

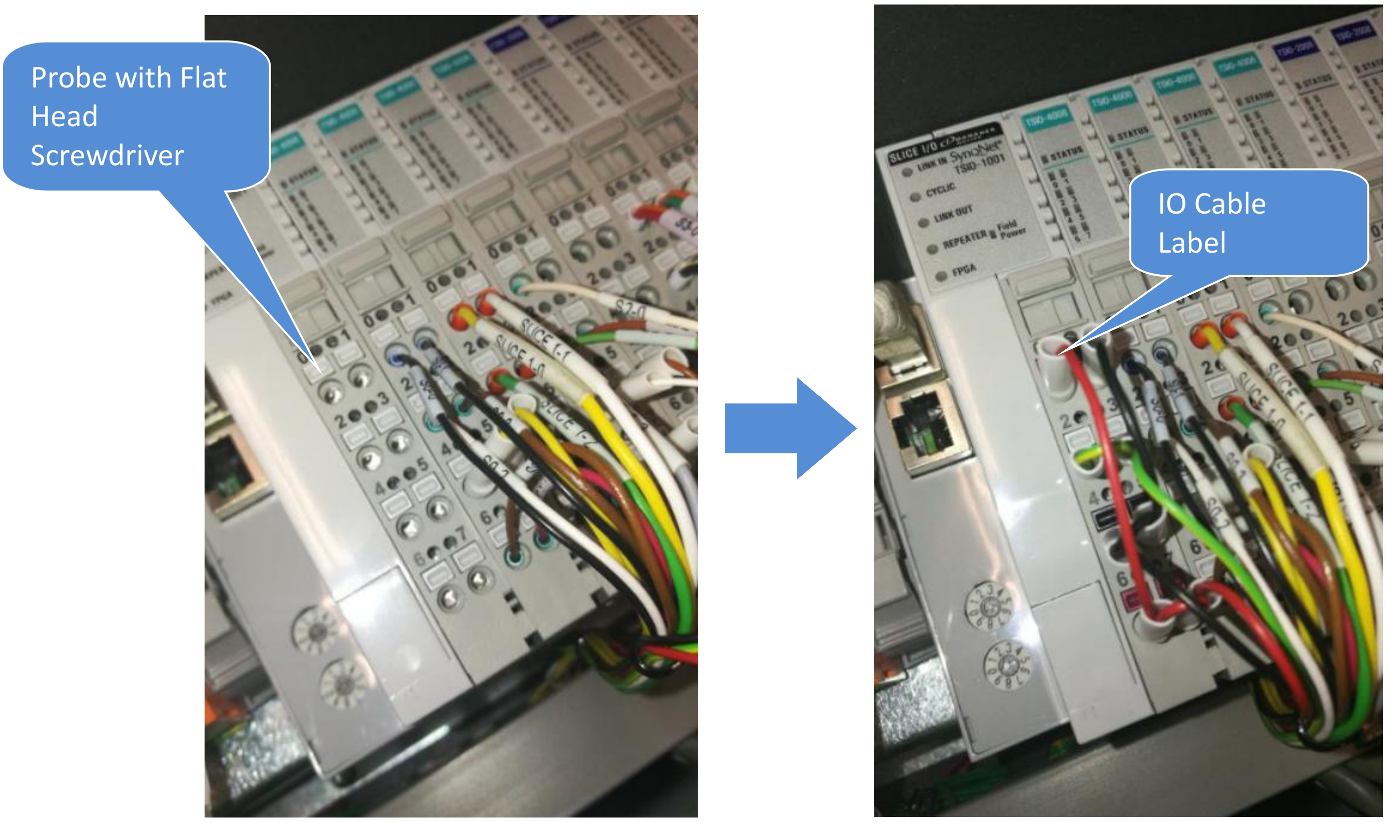

Figure 17 IO Cable Connection

20. Connect LAN cable to 2150-0020 “In” port as shown

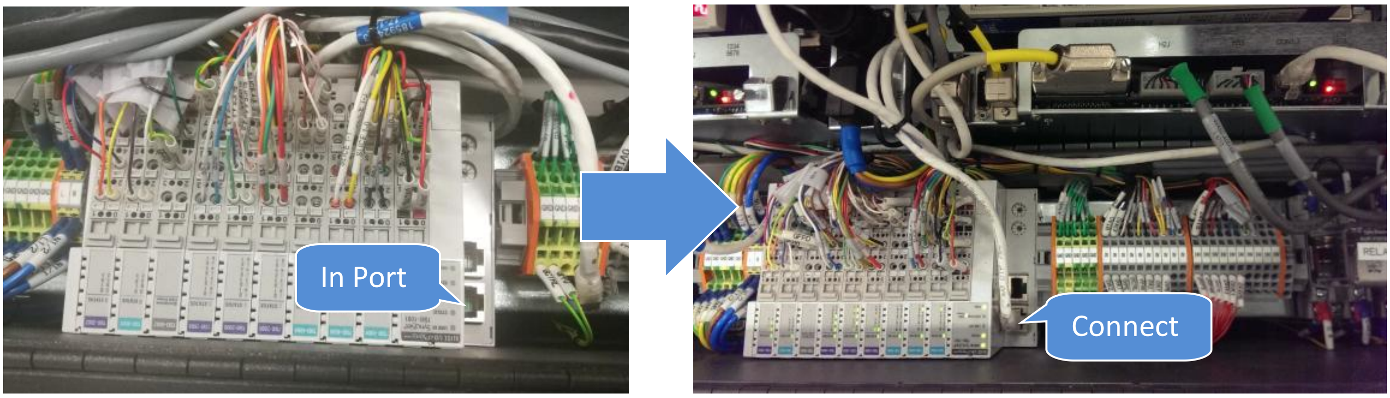

Figure 18 LAN Cable Connection

Selected Slice IO Replacement

21. Prepare Selected Slice IO by refer to Table 1.Unlock the lever.

Figure 19 Selected Slice IO (Reference Picture Only)

22. Replace Selected Slice IO by refer to below orientation.Visual check pins where require firmly mating with Side Slice IO. Gently push in & lock the lever.

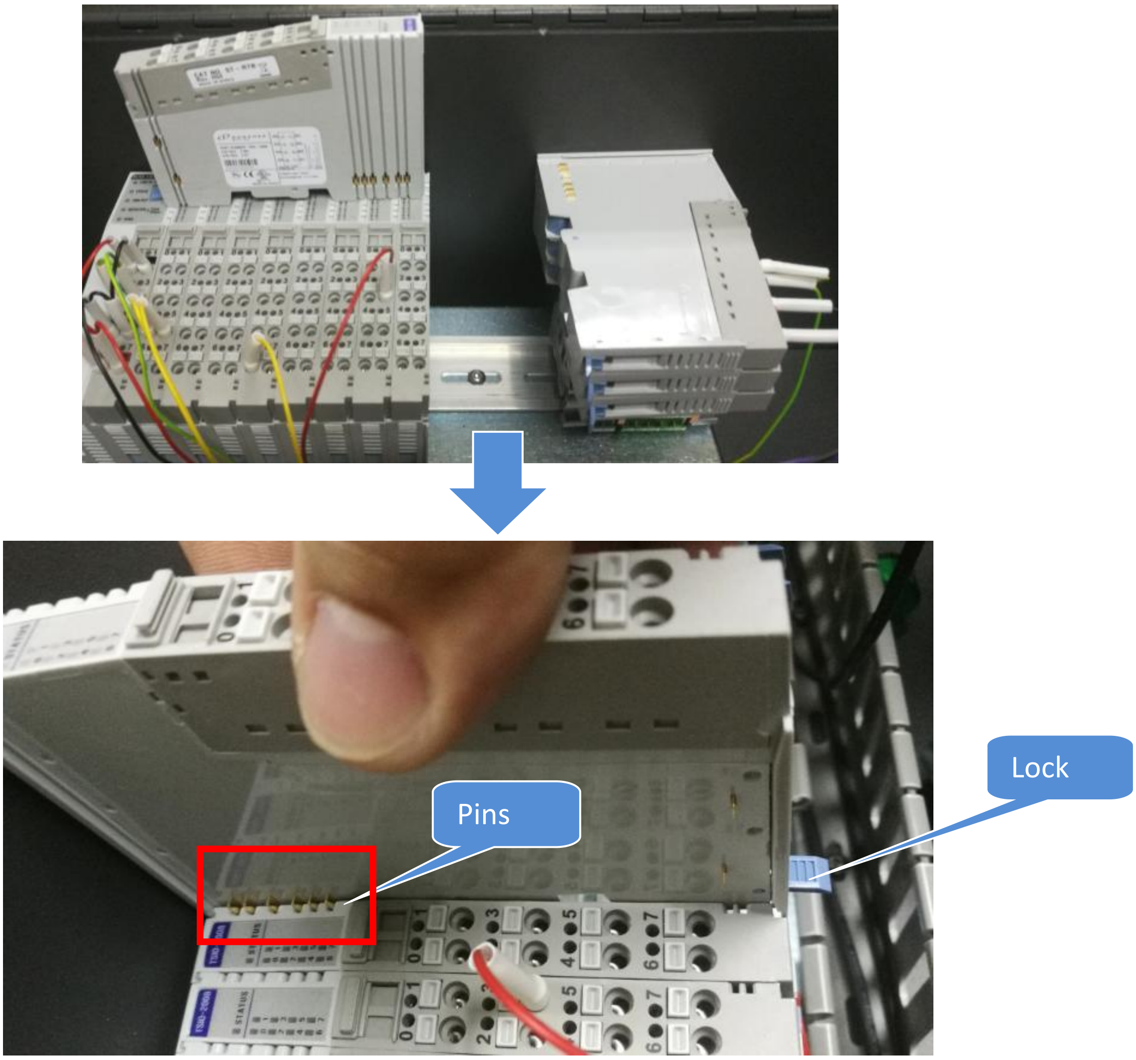

Figure 20

23. Assemble all Side Slice IO respectively by refer to below orientation.Visual check pins where require firmly mating with Side Slice IO. Gently push in & lock the lever.

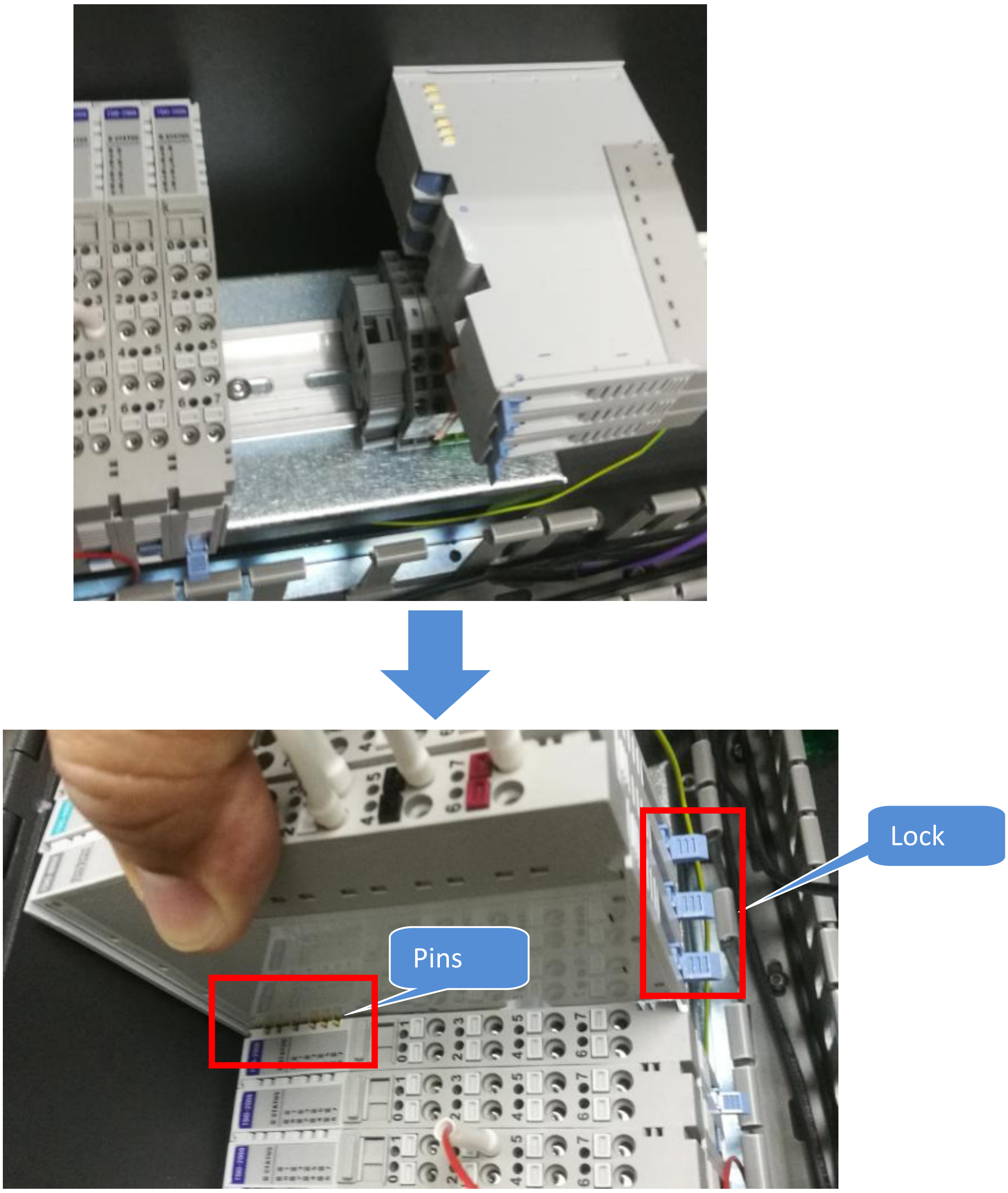

Figure 21 Side Slice IO Assembly

24. Assemble IO Module Side Cover as shown. #Caution on cable snapping upon Side Cover Assembly

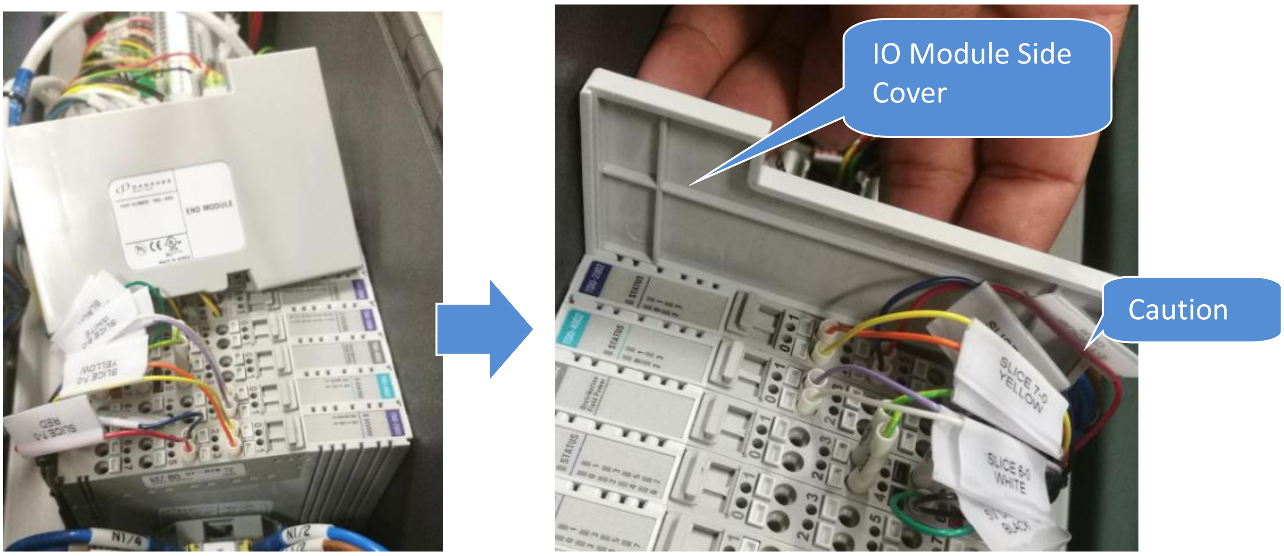

Figure 22 IO Module Side Cover Assembly

25. While probe the Tiny Square Button, gently insert IO Cable respectively by refer to Figure 23

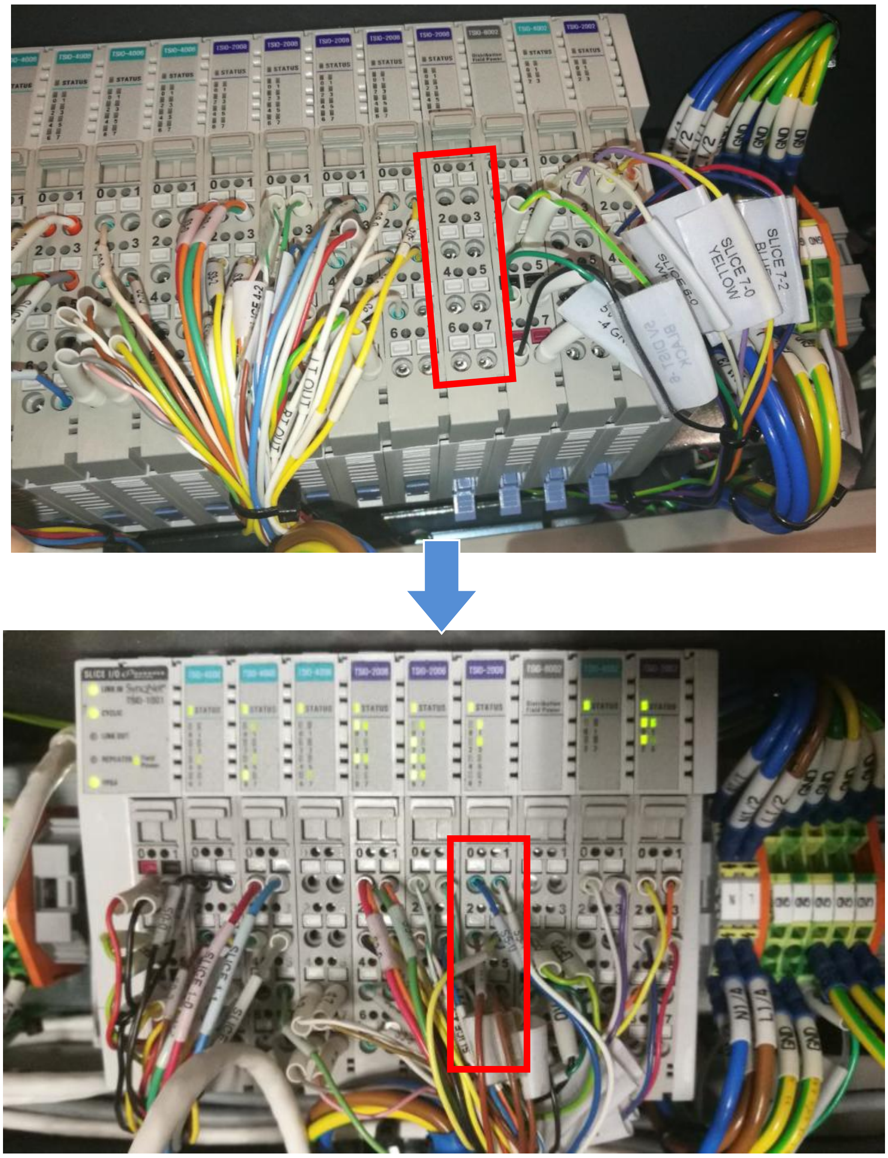

Figure 23

Verification Procedure

26. Use Multi-meter to **probe the 0vDC and 24vDC Terminal block**, Ohm reading should approximately **130 Ohm and above**

    If Ohm Reading Close to Zero , Short Circuit potentially occur . Do Not Power on SSCA. Re-screen all cable connection by refer to Step 18 - 24

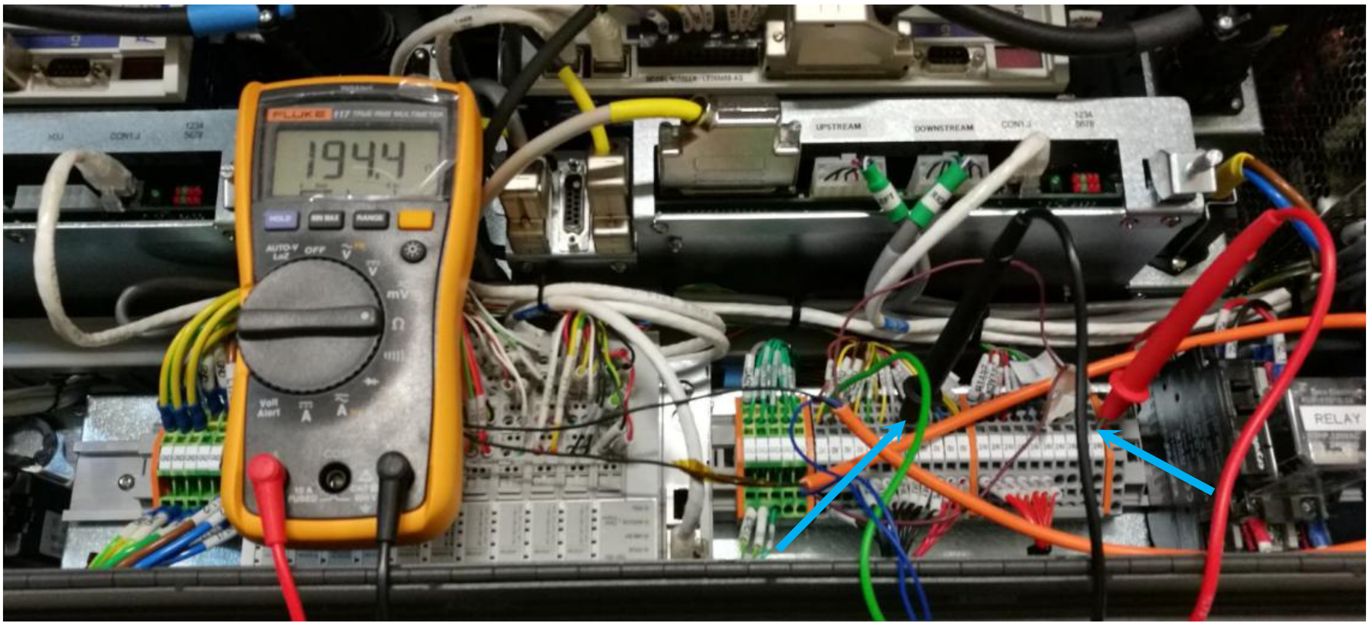

Figure 24 Terminal Block Verification

27. Open SSCA
28. Start-up system as usual
29. Run CDnA